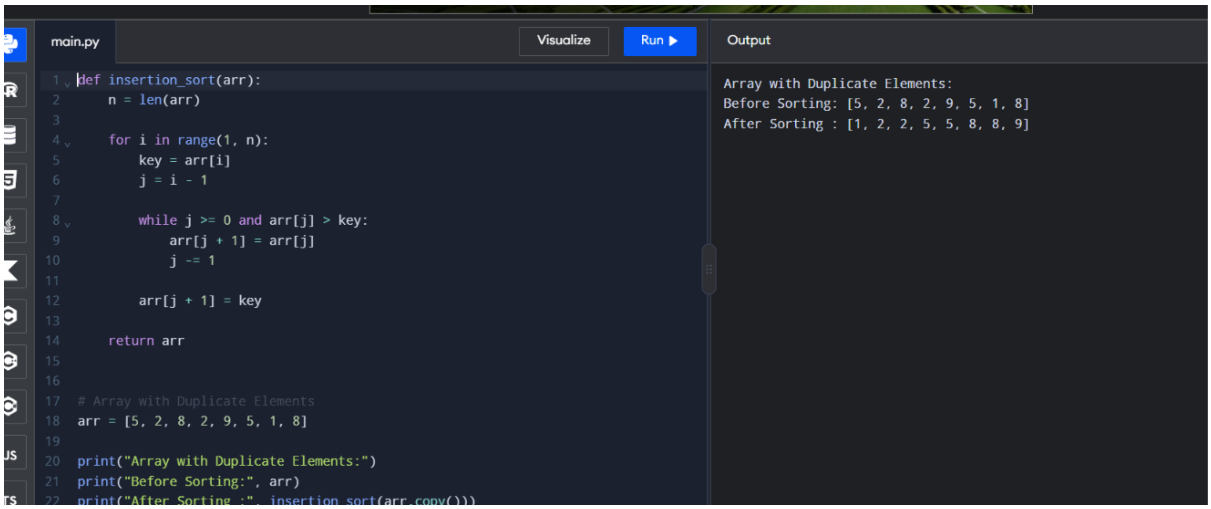
Experiment 8 : Insertion Sort with Duplicate Elements

Aim

To implement the Insertion Sort algorithm and sort an array containing duplicate elements.

Algorithm

1. Consider the first element as already sorted.
2. Take the next element as the key.
3. Compare the key with elements in the sorted portion.
4. Shift all larger elements one position to the right.
5. Insert the key at its correct position.
6. Repeat the process for all remaining elements.
7. Display the sorted array.



Result

The Insertion Sort algorithm was successfully implemented and executed on an array containing duplicate elements. The array was sorted in ascending order while preserving the relative order of duplicate elements.

Time Complexity

Best Case	O(n)
Average Case	$O(n^2)$
Worst Case	$O(n^2)$